MIL-C-82078A(YD)
24 April 1981
SUPERSEDING
MIL-C-82078(YD)
10 June 1966

MILITARY SPECIFICATION

CLAMPS, ELECTRICAL, CONDUCTIVE

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers conductive ground clamping devices for low amperage electrical power and telelphone services.

* 1.2 Classification. Clamps shall be of the following types and sizes (see 6.2):

Type 1 - Adjustable perforated band.

Sizes - 3/8 - 1-1/4 inch.

Type 2 - Nonadjustable.

Sizes - No. 2 American Wire Gage (AWG) - No. 8 AWG.

Type 3 - Adjustable.

Sizes - No. 4 AWG solid - No. 2/0 AWG stranded.

* 1.2.1 Part number. The specification part number for items described in this specification will be identified as shown in 6.3.

* 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposal, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

* U.S. GOVERNMENT PRINTING OFFICE: 1981—703-023/2651

FSC 5935

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SPECIFICATION

MILITARY

MIL-E-117555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 467 - Grounding and Bonding Equipment.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

* 3. REQUIREMENTS

3.1 Standard commercial product. The clamp shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

* 3.2 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

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* 3.2.1 Corrosion protection. Clamps shall conform to UL 467, paragraphs 2.7, 2.7A, and 2.8.

* 3.3 Type 1 clamp. Type 1 clamp shall be of the adjustable type and shall consist of a perforated tinned copper band with locking pin of sufficient size to fit the band perforations, a nut secured to the inside of the band to receive the tightening screw, and a slotted head tightening screw with locknut and washer. A terminal shall be furnished. The band shall be approximately 0.50-inch wide, 0.025-inch thick, and of sufficient length to fit pipes from 3/8- through 1-1/4 inch iron pipe size. When tested in conformance to 4.6, the clamp shall resist a 25-pound pull test.

* 3.4 Type 2 clamp. Type 2 clamp shall be of the nonadjustable type for use with No. 2 AWG to No. 8 AWG solid ground wires when used in conjunction with a 3/4-inch diameter ground rod. Clamp shall be made of copper, copper alloy, or copper clad cast steel and shall consist of a body with a tightening bolt. The bolt shall have an unslotted head with a flat contact point and shall be capable of being screwed into position for tightening purposes. When tested in conformance to 4.6, the clamp shall resist a 25-pound pull test.

* 3.5 Type 3 clamp. Type 3 clamp shall be of the adjustable type for use with No. 4 AWG solid to No. 2/0 AWG stranded wire for attaching to a ground rod of 1-inch diameter or to 1/2-inch or 3/4-inch pipe. Clamp shall be of copper or copper alloy and shall consist of a two-piece cast body, and a U-bolt with nuts and lockwashers. Construction shall be such that the grounded wire or cable may be connected parallel to or at right angles to the ground rod or pipe. When tested in conformance to 4.6, the clamp shall resist a 25-pound pull test.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.2.1).
(b) Packaging inspection (see 4.7).

4.2.1 Quality conformance inspection. Quality conformance inspection shall be performed on the sample units selected in accordance with 4.4. This inspection shall include the examination of 4.5 and the tests of 4.6.

4.3 Inspection lot. All units of the same description offered to the Government at one time, shall be considered a lot for purpose of inspection. The sample shall be one complete unit.

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4.4 Sampling. A random sample of units shall be selected from each lot in accordance with MIL-STD-105.

4.4.1 Sampling for examination. Examination of the units shall be based on inspection level S-4 and an Acceptable Quality Level (AQL) of 2.5 percent defective.

4.4.2 Sampling for tests. Tests of the units shall be based on inspection level S-3 and an AQL of 4.0 percent defective.

4.5 Examination. Each sample selected in accordance with 4.4.1 shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements.

* 4.6 Tests. Each sample clamp selected in accordance with 4.4.2 shall be subjected to a 25-pound pull test. The sample clamp shall be attached to an appropriate cable, pipe, or ground rod and tested for conformance to 3.3, 3.4, or 3.5 as applicable. This test is not intended to supersede or supplant any test conducted by the manufacturer in the fabrication of standard commercial products. Any slippage, deformation, or other failure shall be cause for rejection.

4.7 Packaging inspection. The preservation, packaging, packing, and marking of the clamps shall be inspected to verify conformance to the requirements of Section 5.

* 5. PACKAGING

5.1 Preservation, packaging, packing, and marking. The clamps shall be preserved, packaged, packed, and marked in accordance with MIL-E-17555. The level of preservation and level of packing shall be as specified (see 6.2).

6. NOTES

* 6.1 Intended use. The clamps are intended for ground clamping of low amperage electrical power and telephone services.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Definitive part number of clamp, including type and size required (see 1.2 and 6.3).
(d) Level of preservation, packaging, and level of packing required (see 5.1).

* 6.3 Definitive specification part number. The specification part number is a definitive part number which corresponds to the type and size of clamps

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covered by this specification and defines the requirements of the option presented under this specification. The specification number, the type and size of clamps are combined to form the definitive specification part number.

	M82078	X	X
Military specification number			
Type			
Size			

* 6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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